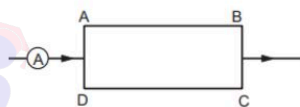


Section : Physics

Q.1

A rectangular loop ABCD made of uniform wire is a part of an electric circuit shown in the given figure. The length $AD = BC = 5 \text{ mm}$. The sides AB and DC are long compared to the other two sides. What will be the magnetic force per unit length acting on the wire DC due to the wire AB if the ammeter reading is 8 A?



1. $1.6 \times 10^{-4} \text{ N m}^{-1}$
2. $6.4 \times 10^{-4} \text{ N m}^{-1}$
3. $3.2 \times 10^{-4} \text{ N m}^{-1}$
4. $6.4 \times 10^{-3} \text{ N m}^{-1}$

Options 1. 1

2. 2
3. 3
4. 4

Q.2

The ratio of the radii of two nuclei of mass numbers A_1 and A_2 respectively would be

1. $(A_1/A_2)^{1/2}$
2. $(A_2/A_1)^{1/3}$
3. $(A_2/A_1)^{1/2}$
4. $(A_1/A_2)^{1/3}$

Options 1. 1

2. 2
3. 3
4. 4

Q.3

In a moving coil galvanometer, the coil is wound over a metallic frame in order to

1. increase moment of inertia
2. reduce hysteresis
3. produce electromagnetic damping
4. increase sensitivity

Options 1. 1

2. 2
3. 3
4. 4

Q.4

A cell of emf 10 V and the internal resistance 1 Ω is being charged by an external supply of 100 V. An external resistance of 8 Ω is connected to the charging circuit. The current in the circuit is

1. 9 A
2. 11 A
3. 12.2 A
4. 10 A

Options 1. 1

2. 2
3. 3
4. 4

Q.5

A long solenoid is formed by winding 40 turns/cm. The current required to produce a magnetic field of 22 mT inside the solenoid will be

(Take $\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$)

1. 2.2 A
2. 4.4 A
3. 0.22 A
4. 0.44 A

Options 1. 1

2. 2
3. 3
4. 4

Q.6

Three capacitors, C_1 , C_2 , C_3 are connected in series. This combination is connected in parallel with capacitor C_4 . $C_1 = C_2 = C_3 = 30 \mu\text{F}$ and $C_4 = 10 \mu\text{F}$. Now a potential 400 V is applied across the combination. The total charge across the combination of capacitors is

1. $8 \times 10^{-3} \text{ C}$
2. $4 \times 10^{-3} \text{ C}$
3. $3.6 \times 10^{-3} \text{ C}$
4. $4 \times 10^{-2} \text{ C}$

Options 1. 1

2. 2
3. 3

- Q.7** The focal length of a lens depends on
1. The radii of curvature of its surfaces and the refractive index of the lens with respect to surrounding medium.
 2. Only the radii of curvature of its surface and the absolute refractive index of its material.
 3. Only the radii of curvature of its surface and the refractive index of the surrounding medium.
 4. Only the radii of curvature of its surface.

Options 1. 1

2. 2
3. 3
4. 4

- Q.8** The phenomenon of polarisation of light confirmed that light waves are
1. transverse in nature
 2. longitudinal in nature
 3. electromagnetic in nature
 4. particles in nature

Options 1. 1

2. 2
3. 3
4. 4

- Q.9** Eg (energy band-gap) for silicon is greater than that for germanium. Therefore, it can be concluded that
1. a greater number of electron hole pairs will be generated in silicon than in germanium at room temperature
 2. less number of electron hole pairs will be generated in silicon than in germanium at room temperature
 3. an equal number of electron hole pairs will be generated in both at lower temperatures
 4. an equal number of electron hole pairs will be generated in both at higher temperatures

Options 1. 1

2. 2
3. 3
4. 4

- Q.10** In the depletion region of an unbiased p-n junction diode there are
1. mobile ions
 2. immobile ions
 3. electrons
 4. holes

Options 1. 1

2. 2
3. 3
4. 4

- Q.11** The desirable properties of core of a transformer are
1. Low permeability and low hysteresis loss
 2. High permeability and High hysteresis loss
 3. High permeability and low hysteresis loss
 4. Low permeability and high hysteresis loss

Options 1. 1

2. 2
3. 3
4. 4

- Q.12** Identify the correct statements in the following.
- (A) Optical fibres consist of a core and cladding
- (B) Optical fibres are fabricated such that light reflected at one side of the inner surface strikes the other at an angle smaller than the critical angle.
- (C) Optical fibres make use of the phenomenon of total internal reflection.
- (D) Optical fibres can be used as an optical pipe
- Choose the correct answer from the options given below:
1. (A), (B) and (C) only
 2. (A), (C) and (D) only
 3. (B), (C) and (D) only
 4. (A) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

- Q.13** In Young's double slit experiment, the width of the fringes obtained with light of wavelength $6000 \text{ \AA}$ is 3.0 mm . If the entire apparatus is immersed in a liquid medium of refractive index 1.50 , then the fringe width would be
1. 1.3 mm
 2. 2.0 mm
 3. 1.2 mm
 4. 1.5 mm

Options 1. 1

2. 2
3. 3
4. 4

- Q.14** A real and smaller image of an object will be produced by
1. a concave mirror, when the object is placed beyond $2F$
 2. a convex mirror, when the object is placed between the Pole and the principal focus F
 3. a concave mirror, when the object is placed between the center of curvature (R) and F
 4. a convex mirror, when the object is placed beyond $2F$

Options 1. 1

2. 2

3. 3

4. 4

Q.15 Identify the correct statements among the following.

(A) For two unlike equal charges, there is a neutral point on the middle point on the line joining the two charges.

(B) For two, unlike unequal charges, the neutral point will be outside on the line joining the two charges.

(C) For two positive equal charges, the neutral point will be at the middle point of the line joining the two charges.

(D) For two negative charges placed along a line, there will not be any neutral point.

Choose the correct answer from the options given below:

1. (A) and (B) only

2. (A) and (C) only

3. (B) and (C) only

4. (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.16 A needle of length 4.5 cm is placed vertically 12 cm away from a convex mirror of focal length 15 cm. The magnification of the image produced is

1. -5/9

2. 5/9

3. 20/9

4. 10/9

Options 1. 1

2. 2

3. 3

4. 4

Q.17 If the potential difference across a capacitor is changed from 12 V to 24 V, the work done is W. How much work will be done if the potential difference is changed from 24 V to 48 V?

1. 4 W

2. 3 W

3. 2 W

4. W

Options 1. 1

2. 2

3. 3

4. 4

Q.18 The magnetic force can act on a charge, which is

1. moving in the perpendicular direction to the magnetic field

2. moving in the direction of the magnetic field

3. moving in the direction opposite to the magnetic field

4. at rest

Options 1. 1

2. 2

3. 3

4. 4

Q.19 Arrange the following waves in the increasing order of their wavelengths

(A) Visible light waves

(B) Microwave waves

(C) Infrared waves

(D) AM radio waves

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)

2. (A), (C), (B), (D)

3. (B), (A), (D), (C)

4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.20 A very long straight wire carries a current (I) directed along positive Y-axis. A point charge +Q moves with velocity (v) along X -axis. The direction of force on the charge at point (5,0,0) m is

1. along positive x-axis

2. along positive y-axis

3. along negative z-axis

4. along positive z-axis

Options 1. 1

2. 2

3. 3

4. 4

Q.21 A current of 2 A flows through a $2\ \Omega$ resistor when connected with a cell. When the resistance is replaced by $10\ \Omega$, a current of 0.5 A flows. The internal resistance of the cell is

1. $1/3\ \Omega$

2. $1\ \Omega$

3. $3/2\ \Omega$

4. $2/3\ \Omega$

Options 1. 1

2. 2
3. 3
4. 4

Q.22 If the half-life of a radioactive nuclide is 16 hours, what fraction of original sample will remain undecayed after 48 hours ?

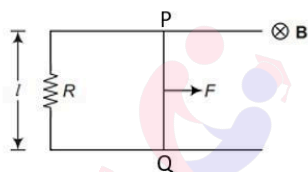
1. $\frac{1}{2}$
2. $\frac{1}{4}$
3. $\frac{1}{8}$
4. $\frac{1}{16}$

Options 1. 1

2. 2
3. 3
4. 4

Q.23

In the given figure a rod PQ of length 2.5 m can slide without friction. The value of R is 5.0Ω and a magnetic field of 2 T is directed perpendicularly into the paper. The force F required to move the rod to the right at a constant speed of 1.0 m/s will be



1. 2.5 N
2. 5 N
3. 10 N
4. 1 N

Options 1. 1

2. 2
3. 3
4. 4

Q.24 When a positive charge (+ q) is brought near one face of a metal cube, the

1. cube becomes negatively charged
2. cube becomes positively charged
3. face near the charge (+ q) becomes positively charged, and the opposite face becomes negatively charged
4. face near the charge (+ q) becomes negatively charged, and the opposite face becomes positively charged

Options 1. 1

2. 2
3. 3
4. 4

Q.25 In a Wheatstone Bridge, all the four arms have equal resistance r. If the resistance of the galvanometer arm is also equal to r, the equivalent resistance of the combination will be

1. $0.25 r$
2. $0.5 r$
3. r
4. $2 r$

Options 1. 1

2. 2
3. 3
4. 4

Q.26 What will be the momentum of a photon of energy 0.1 MeV in SI unit?

1. 5.33×10^{-22}
2. 7×10^{-24}
3. 5.33×10^{-23}
4. 3.33×10^{-23}

Options 1. 1

2. 2
3. 3
4. 4

Q.27 If the intensity of incident radiation on metal is doubled, the kinetic energy of electrons emitted would be

1. doubled
2. halved
3. unaffected
4. four times

Options 1. 1

2. 2
3. 3
4. 4

Q.28 The torque acting on a magnet of magnetic length 8 cm and pole strength 20 A m placed in a magnetic field of 3×10^{-5} T at an angle of $\pi/6$ rad will be

1. 2.4×10^{-4} Nm
2. 2.4×10^{-5} Nm
3. 4.2×10^{-3} Nm
4. 4.2×10^{-5} Nm

Options 1. 1

2. 2
3. 3
4. 4

Q.29 A steady current flows in a metallic conductor of non-uniform cross-section. The quantity that is constant along the length is

1. Both current and electric field
2. Drift speed only
3. Both current and drift speed
4. Current only

Options 1. 1

2. 2
3. 3
4. 4

Q.30 The velocity of light in a medium depends on

1. electric and magnetic properties of the medium
2. electric properties of the medium only
3. magnetic properties of the medium only
4. the permittivity of free space only

Options 1. 1

2. 2
3. 3
4. 4

Q.31 An object of length 2.0 cm is placed at a distance of 1.5 f from a concave mirror where f is the magnitude of the focal length of the mirror. The length of the object is perpendicular to the principal axis. The length of the image will be

1. 4.0 cm
2. -4.0 cm
3. 2.0 cm
4. -2.0 cm

Options 1. 1

2. 2
3. 3
4. 4

Q.32 When a metal rod of length L placed normal to a uniform magnetic field B is moved at a velocity of v perpendicular to the field, the induced emf across its ends is

1. 0
2. BLv
3. BL/v
4. (BLv)/2

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Identify the incorrect statements among the following.

- (A) The two charges of an electric dipole will experience force in the same direction when placed in a uniform electric field,
- (B) The dipole moment is a vector quantity,
- (C) The direction of the dipole moment is from +q charge towards -q charge,
- (D) The torque acting on the dipole in an electric field depends on the intensity of the field and the dipole moment.

Choose the correct answer from the options given below:

1. (B) and (C) only
2. (C) and (D) only
3. (B) and (D) only
4. (A) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.34 A parallel plate capacitor of capacitance $2 \mu\text{F}$ has energy 10×10^{-5} J The potential difference and the charge on the capacitor are:

1. 20 V, $10 \mu\text{C}$
2. 100 V, $200 \mu\text{C}$

3. 10 V, 20 μC
4. 3.16 V, 9.5 μC

Options 1. 1
2. 2
3. 3
4. 4

Q.35 The shortest wavelength in the Balmer series of hydrogen atom would be
(Given $R = 1.1 \times 10^7 \text{ m}^{-1}$)

1. 3636 Å
2. 4529 Å
3. 2433 Å
4. 3454 Å

Options 1. 1
2. 2
3. 3
4. 4

Q.36 In a Young's double slit experiment, two slits are separated by 3 mm and the screen is placed 1.5 m away. The distance between two consecutive minima for a light of wavelength 420 nm is

1. 0.84 mm
2. 0.63 mm
3. 0.42 mm
4. 0.21 mm

Options 1. 1
2. 2
3. 3
4. 4

Q.37 The electric potential for an electric dipole

1. varies inversely with the distance from the center of the dipole on the equatorial line.
2. is a vector quantity
3. is zero at mid-point on the line, joining two charges of dipole
4. has a direction from charge -q towards +q

Options 1. 1
2. 2
3. 3
4. 4

Q.38 A set of n equal resistors, each of resistance R are connected in series to a battery of emf E and internal resistance R. The current drawn is I. Now, the n resistors are connected in parallel to the same battery. The current drawn from the battery is 20 I. The value of n is

1. 10
2. 11
3. 20
4. 21

Options 1. 1
2. 2
3. 3
4. 4

Q.39 A 50 cm long air core solenoid has a 2 cm radius and 500 turns. The self-inductance of the solenoid would be
(given $\mu_0 = 4\pi \times 10^{-7} \text{ Tm} \cdot \text{A}^{-1}$ and $\pi^2 = 10$)

1. $5.0 \times 10^{-4} \text{ H}$
2. $6.0 \times 10^{-4} \text{ H}$
3. $8.0 \times 10^{-4} \text{ H}$
4. $0.1 \times 10^{-4} \text{ H}$

Options 1. 1
2. 2
3. 3
4. 4

Q.40 The relative motion between the magnet and a coil is responsible for generation (induction) of electric current in the coil. This is in accordance with

1. Gauss's law
2. Biot-Savart's law
3. Ampere's Circuital Law
4. Faraday's Law

Options 1. 1
2. 2
3. 3
4. 4

Q.41 Identify the incorrect statement(s) from the following in the context of an electric current flowing through a conductor.

- (A) The electrons suffer collisions with heavy fixed ions in the conductor
(B) The resistance of the given conductor depends on its temperature
(C) Resistance and resistivity both depend on the dimensions of the conductor
(D) When a potential difference V is applied across the conductor then $V = IR$
Choose the correct answer from the options given below:

1. (A) and (C) only
2. (B) only
3. (C) only
4. (B) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.42 In a radioactive decay, neither the atomic number nor the mass number changes. Which of the following is emitted in the decay?

1. Proton
2. Neutron
3. Electron
4. Photon

Options 1. 1
2. 2
3. 3
4. 4

Q.43 Electromagnetic waves are produced by

1. charge at rest
2. charge in uniform motion
3. charge particle moving along a circular path
4. charge particle moving with velocity v along the direction of magnetic field

Options 1. 1
2. 2
3. 3
4. 4

Q.44 The reverse biasing in a p-n junction diode

1. decreases the potential barrier height
2. increases the potential barrier height
3. decreases the number of minority charge carriers
4. increases the number of majority charge carriers

Options 1. 1
2. 2
3. 3
4. 4

Q.45 The ratio of the de-Broglie wavelengths associated with protons accelerated through a potential of 128 V and alpha particles accelerated through a potential of 64 V will be

1. 1:2
2. 1:4
3. 2:1
4. 4:1

Options 1. 1
2. 2
3. 3
4. 4

Q.46 A point charge of $17.7 \mu\text{C}$ is located at the center of a cube of side 0.03 m. Find the electric flux through each face of the cube.

1. $3.3 \times 10^5 \text{ Nm}^2 \text{ C}^{-1}$
2. $1.3 \times 10^5 \text{ Nm}^2 \text{ C}^{-1}$
3. $5.3 \times 10^5 \text{ Nm}^2 \text{ C}^{-1}$
4. $6.3 \times 10^5 \text{ Nm}^2 \text{ C}^{-1}$

Options 1. 1
2. 2
3. 3
4. 4

Q.47 Some of the commonly known ideas regarding magnetism are:

- (A) It is possible to make magnets out of iron and its alloys
- (B) It is possible to isolate the north or south pole of a magnet
- (C) A bar magnet suspended freely, points in the north-south direction
- (D) The earth behaves as a magnet with the magnetic field pointing approximately from the geographic south to the north

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (C) only
3. (B), (C) and (D) only
4. (A), (B), (C) and (D)

Options 1. 1
2. 2
3. 3
4. 4

Q.48 The magnetic field at the centre of a closely wound circular coil of 100 turns, radius 22 cm, carrying a current of 3.5 A will be

1. 1×10^{-2} T
2. 2×10^{-3} T
3. 1×10^{-3} T
4. 2×10^{-4} T

Options

1. 1
2. 2
3. 3
4. 4

Q.49 By increasing the temperature, resistivity of a conductor and a semiconductor

1. decreases for a conductor and increases for a semiconductor
2. increases for a conductor and decreases for a semiconductor
3. decreases for both
4. increases for both

Options

1. 1
2. 2
3. 3
4. 4

Q.50 An LCR series circuit consisting of resistance 45Ω , inductive reactance 4Ω and capacitive reactance 4Ω is connected to an alternating source of emf 90 V. The readings of the ammeter connected to the circuit would be

1. 2 A
2. 1 A
3. $2\sqrt{2}$ A
4. 0 A

Options

1. 1
2. 2
3. 3
4. 4